

GRA 1

GRA 1

Graph A

10 February year

1) GRA 1 - Page 30

2) Indications are on the graph

$$\begin{aligned} m &= \frac{\Delta y}{\Delta x} \\ &= \frac{35}{47} \\ &= 0,74 \Omega / ^\circ\text{C} \end{aligned}$$

3)

$$y = mx + c$$

$$r_t = 0,74(\Omega/^\circ\text{C})t + 149,5$$

$$\text{equation } r = 0,74(\Omega/^\circ\text{C})t + 149,5$$

$$y = mx + c$$

4i)

$$r = 0,74(5) + 149,5$$

$$= 152,7 \Omega \quad \text{read from graph}$$

$$y = mx + c$$

$$y = 0,74(0) + c$$

$$158 = 0,74(11,4) + c$$

$$158 = 8,489 + c$$

$$c = 149,5 \text{ units?}$$

ii)

$$y = mx + c$$

$$r = 0,74(25) + 149,5$$

$$= 167,5 \Omega$$

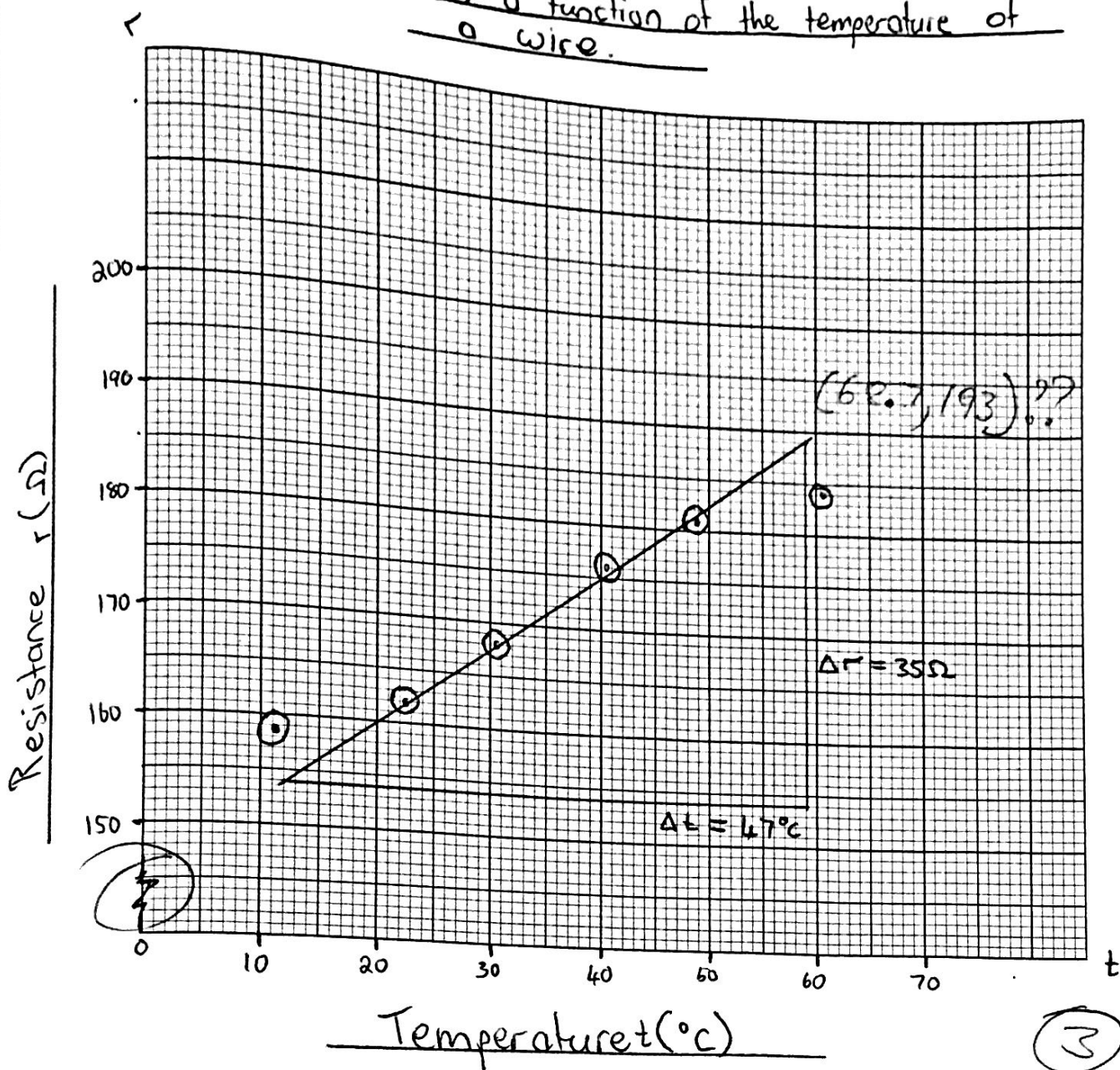
read from graph

GRA 1

DATE: 10 February

EKSP/EXP: GRA 1

The resistance as a function of the temperature of a wire.



GRA 1

Gra 1 Graph B

10 February year

1) GRA 17 - last page

$$2) n = \frac{\log(a_2) - \log(a_1)}{\log(T_2) - \log(T_1)} \quad \textcircled{1}$$

$$= \frac{\log\left(\frac{a_2}{a_1}\right)}{\log\left(\frac{T_2}{T_1}\right)}$$

$$= \log\left(\frac{19.1}{0.386}\right)$$

$$\log\left(\frac{84}{0.241}\right)$$

$$n = 0.665... \quad \textcircled{1}$$

3) You can't take the log because log of 1 is zero and if the constant is equal to zero all the answers will be zero not really $\textcircled{1}$ $a = kT^n$ then $a = T^n$ iff $k=1$ constant $\approx k=1$ ✓

$$4) a = T^{0.66}$$

$$a(a.u.) = T(yr)^{0.66} \quad \textcircled{2}$$

5) Yes it does because with the $\log^x - \log$ graph we can see that the semi-major axes of the orbits is directly proportional to the time it takes to complete an orbit. $\textcircled{1}$

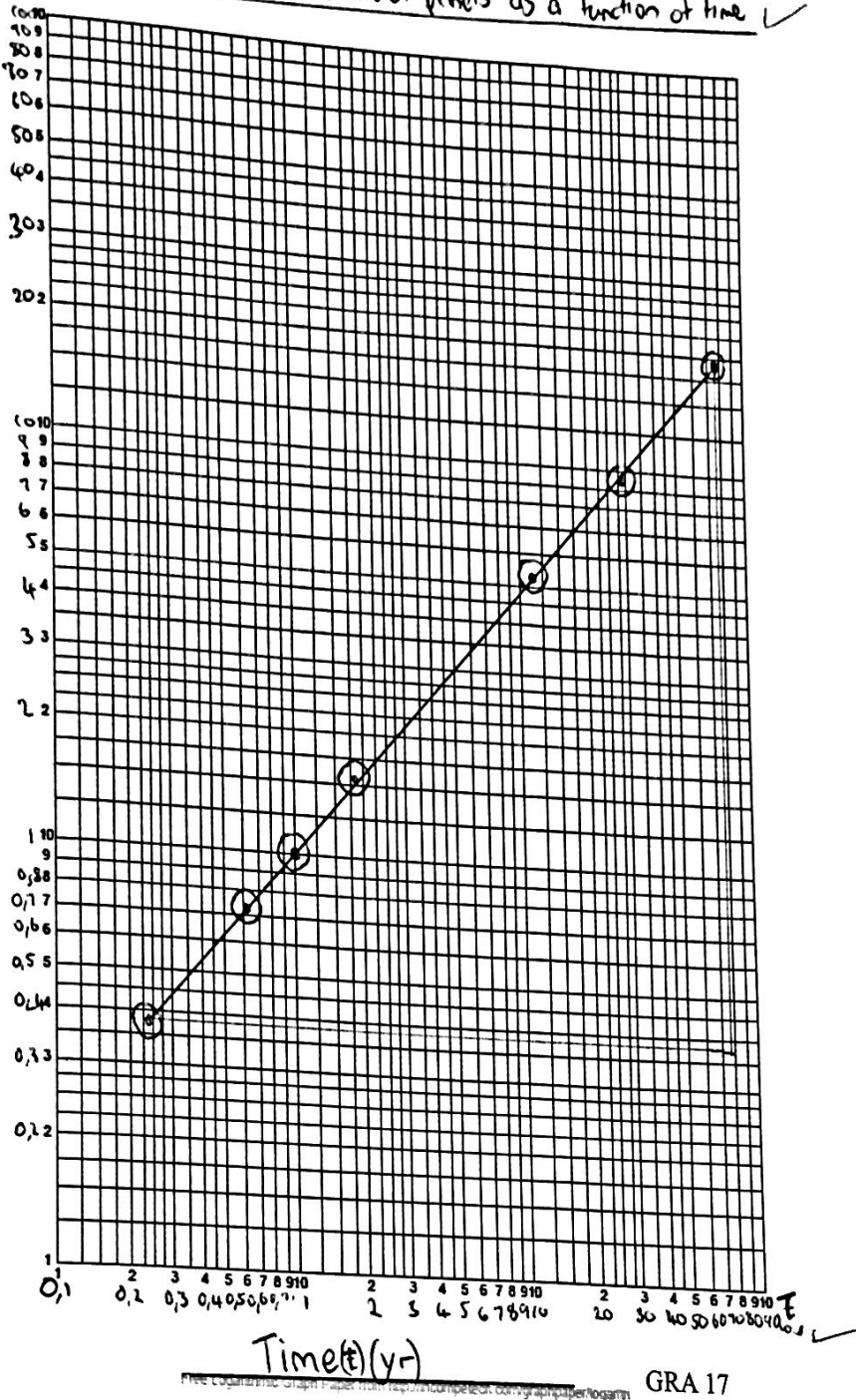
GRA 1

DATE: 10 February.....

EKSP/EXP: (GRA.1)

Semi-major axes of orbits of planets as a function of time ✓

Semi-major axes of the
Orbits of the planets (a)



Free Logarithmic Graph Paper with horizontal and vertical grid lines

GRA 17

4

nice

GRA 2

2015-02-17

GRA 2

1) on GRA 2

2)	x Time (min)	y ln (count rate)
	0,0	6,68
	5,0	5,99
	10,0	5,32
	15,0	4,65
	20,0	3,91

3) on GRA 3

4) $y_0 = 6,68$ ①

$$\alpha = \frac{y_2 - y_1}{x_2 - x_1}$$

$$= \frac{4,1 - 6,2}{18,75 - 3,75}$$

$$= -0,14 \frac{\text{min}^{-1}}{\text{min}} \quad \text{②}$$

5) $\ln(R) = 6,68 - 0,14t$ ③

$\ln(R) = 6,68 - 0,14 \text{ min}^{-1}t$ wrong units

6) $y_0 = \ln(R_0)$ ④

$$6,68 = \ln(R_0)$$

$$\therefore R_0 = e^{6,68} \text{ units}$$

$$\alpha = -\lambda$$

$$-0,14 = -\lambda \quad \text{⑤}$$

$$\lambda = 0,14 \text{ min}^{-1}$$

7) $\ln(R) = \ln(e^{6,68}) + 0,14 \text{ min}^{-1}t$ X

Gro 2

2015 - 02 - 17

$$8) \ln(2) = -0,14 \text{ min}^{-1} T_{\frac{1}{2}} \quad \times$$

$$9) - \frac{\ln(2)}{0,14 \text{ min}^{-1}} = T_{\frac{1}{2}}$$

$$\frac{-\ln 2}{0,14 \text{ min}^{-1}} = T_{\frac{1}{2}}$$

$$T = \frac{-\ln 2}{0,14}$$

$$= -4,95 \text{ min}$$

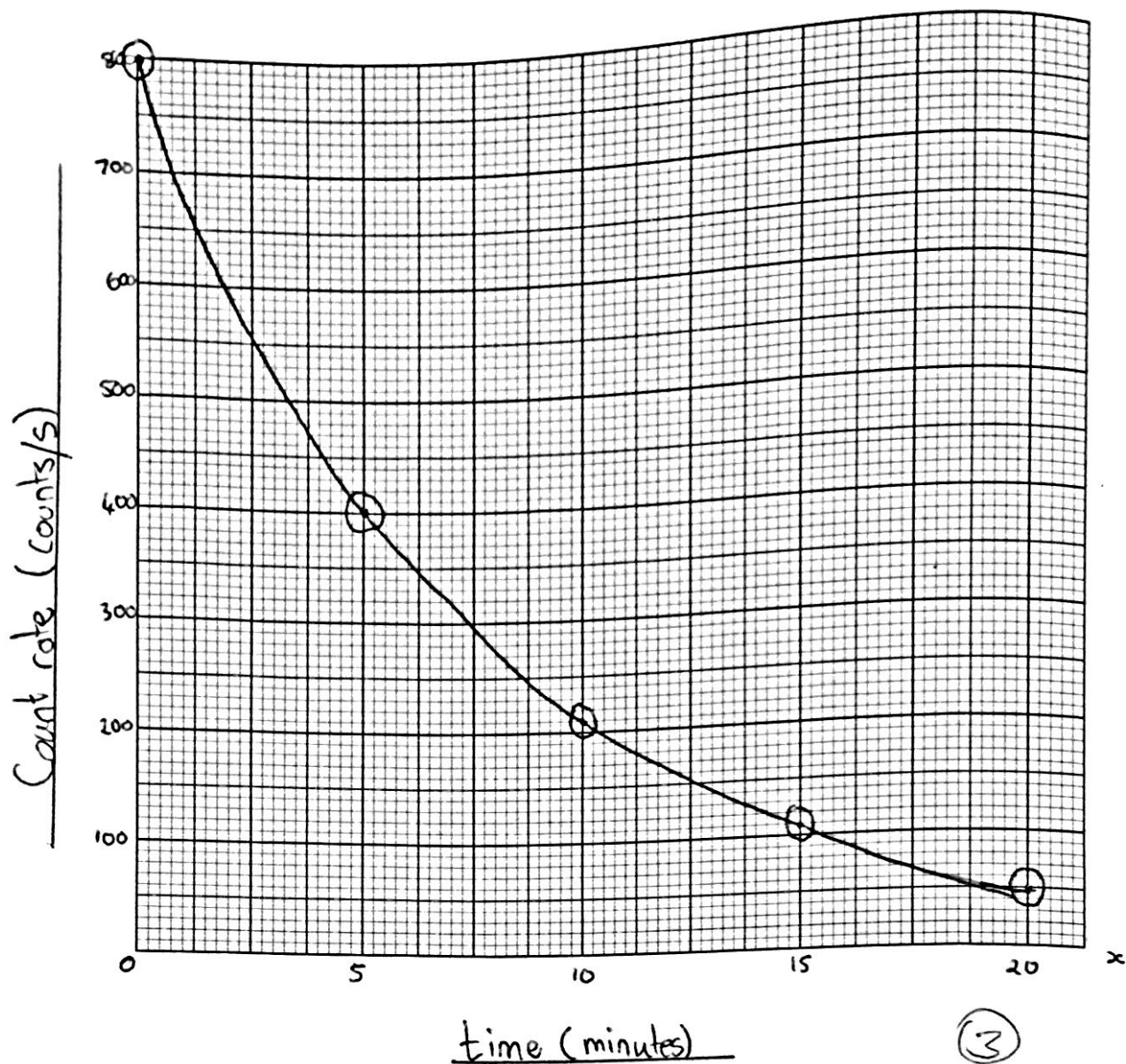
①

GRA 2

EKSP/EXP: ... GRA 2

DATE: 17 February

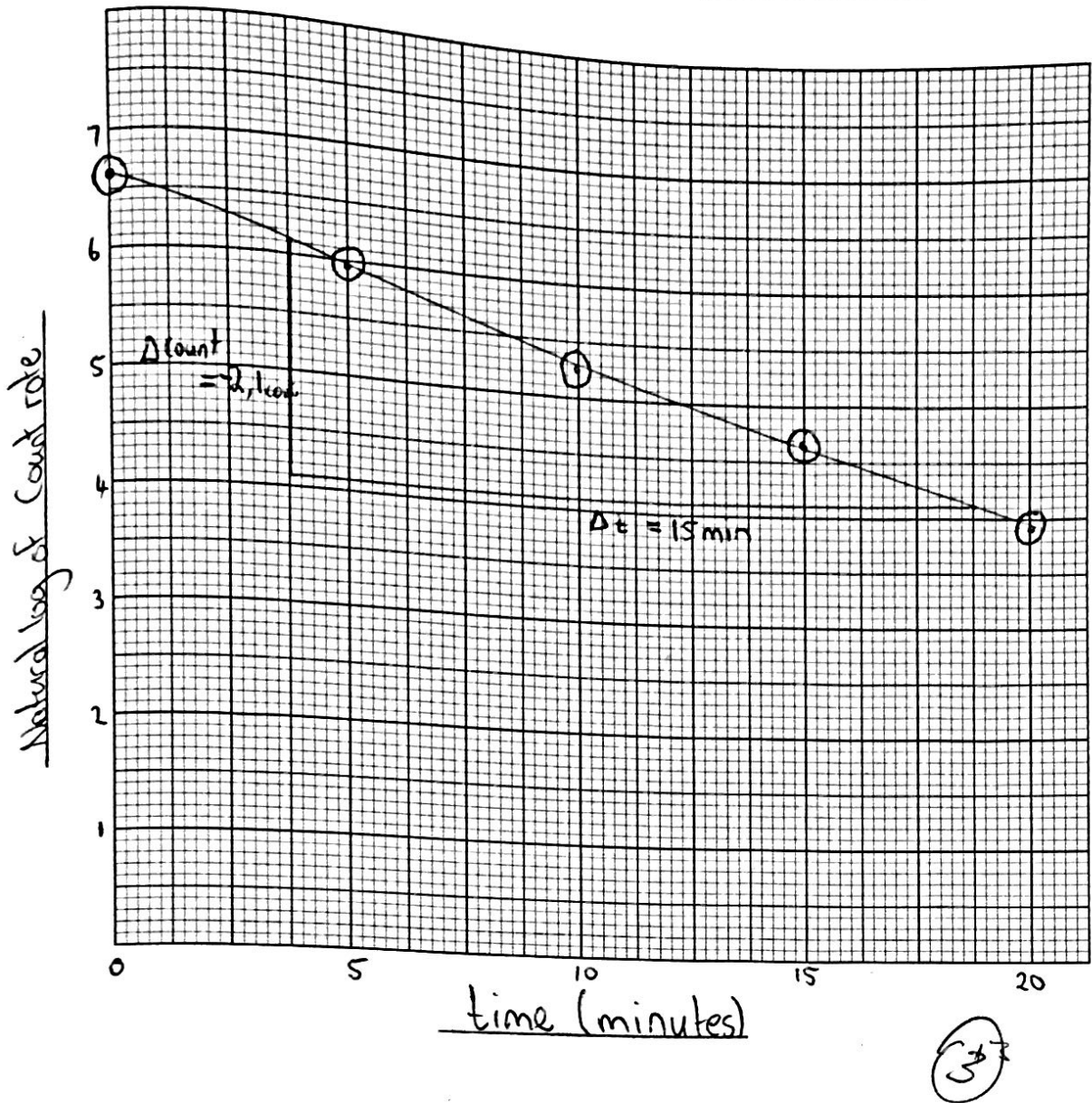
Count rate as a function of time



DATE: 17 February

EKSP/ EXP: GRA 2

Natural log of count rate as a function of time



14/20. (1)

3 March 2015

Eks

Aim: To obtain some insight into the mathematical description of an exponentially decaying process

1) (1)

T	# of remaining cubes
0	100
1	80
2	62
3	44
4	35
5	28
6	167
7	11
8	5
9	5
10	4
11	4
12	3
13	2
14	1
15	1

(2)

Data analysis

2) GRA4

3) $T_{1/2} = 2.6$ Throws from 100 to 50
 $: 4.9$ Throws from 50 to 25

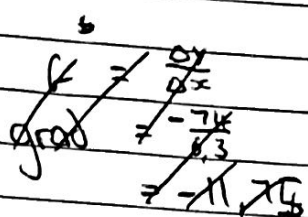
$T_{1/2}$ average: 3.75 Throws

it is closer to 4 so one could round it to 4 throws if you need to. (1)

4) GRAF IS

5) Yes they did, the log graph shows exponential as linear.

6) ~~revised~~



$$\begin{aligned}
 G &= \frac{\Delta y}{\Delta x} \\
 &= \frac{\log(6) - \log(74)}{6.3} \\
 &= \frac{-1.04}{6.3} \\
 &= -0.173
 \end{aligned}$$

Decay constant $\rightarrow 0.173$

$$\gamma = \frac{\text{grad}}{\log e}$$

7) $n = n_0 e^{-\lambda t}$

$$50 = 100 e^{-0.173t}$$

$$\frac{1}{2} = e^{-0.173t}$$

$$\ln\left(\frac{1}{2}\right) = -0.173t$$

$$-0.69 = -0.173t$$

$$4.00 = T_h$$

$$T_h = 4 \text{ throws}$$

8) ~~the calculated value will be more accurate, it has been completed using raw data and not by looking at a graph~~ the graphically represented one because it is raw data that was experimented.

9) 1 in 6 chance

10) 1 throw provide they all land on the right side

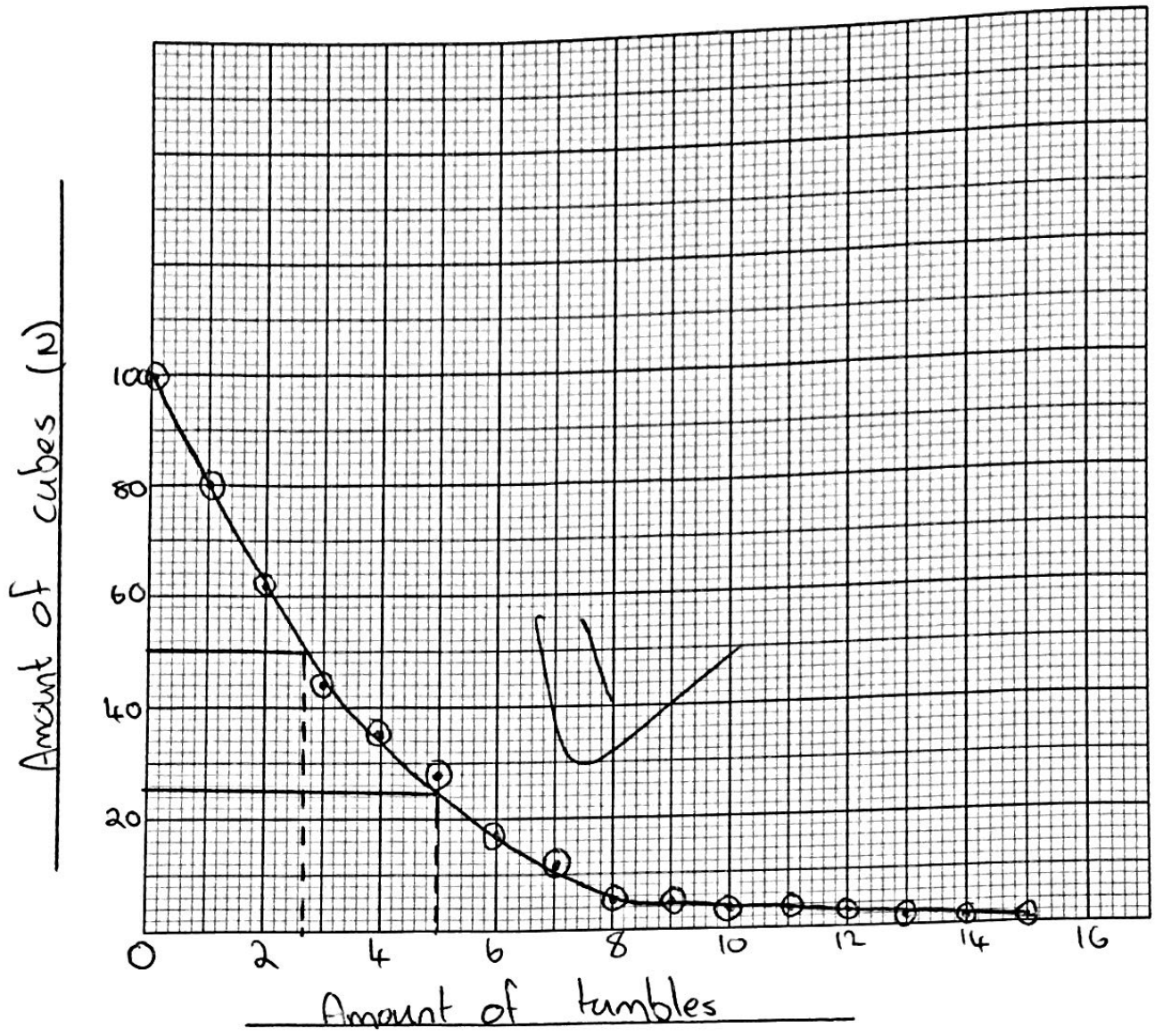
11) 97.66 nuclei

12)

DATE: 3 March

EKSP/EXP: ...EKS...

Amount of cubes remaining as a function of tumbles



17/03/15

PAR

Aim:

To determine the path or trajectory of a horizontally launched projectile and the relationship between the co-ordinates of its trajectory

Results : Determining V_0

1) Point	Horizontal dis (m)
1	2.81
2	2.86
3	2.87
4	2.93
5	2.96
Average	2.89

2) ball radius = ~~10 mm~~ ^{13.75 mm}

3) Height = 0.86m

$$y = y_0 - \frac{g}{2V_0^2} x^2$$

$$-0.86m = \frac{-9.81}{2V_0^2} (2.89)^2$$

$$2V_0^2(0.86) = 9.81(2.89)^2$$

$$2V_0^2 = 95.27$$

$$V_0 = 6.90 \text{ m.s}^{-1}$$

Co-ordinates of the ball along the trajectory

Horizontal distance	Vertical distance	$x^2 \text{ m}^2$
0,58 m	0,85 m	0,34 m
1,15 m	0,74 m	1,32 m
1,71 m	0,58 m	2,92 m
2,28 m	0,325 m	5,20 m
2,89 m	0 m	8,35 m

1) the co-ordinate system would stay the same ✓

Graph on GRA 8

Question 2

it forms a straight line ^{contradicting yourself parabolic} therefore its a ~~linear~~ relationship

3) $y = mx + c$

$y = mx + 0,85$

$0,74 = m(1,32) + 0,85$

$-\frac{11}{100} = m(1,32)$

$m = -\frac{1}{12}$

$y = -0,08x + 0,86$

$y - 0,12 m = -0,11(x - 3)$

$y = -0,11x^2 + 0,55m$

4) $m = \frac{y_2 - y_1}{x_2 - x_1}$
 $= \frac{0,325 - 0,85}{5,20 - 0,34}$
 $= -0,108 \text{ units}$

$-\frac{g}{2v_0^2} = -0,108$

$\frac{9,81}{0,108} = 2v_0^2$

$v_0 = 6,74 \text{ m.s}^{-1}$

5) $\% \text{ difference} = \frac{6,74}{6,90} \times 100 = 97,69\%$

therefore $\% \text{ difference} = 2,31\% \checkmark$

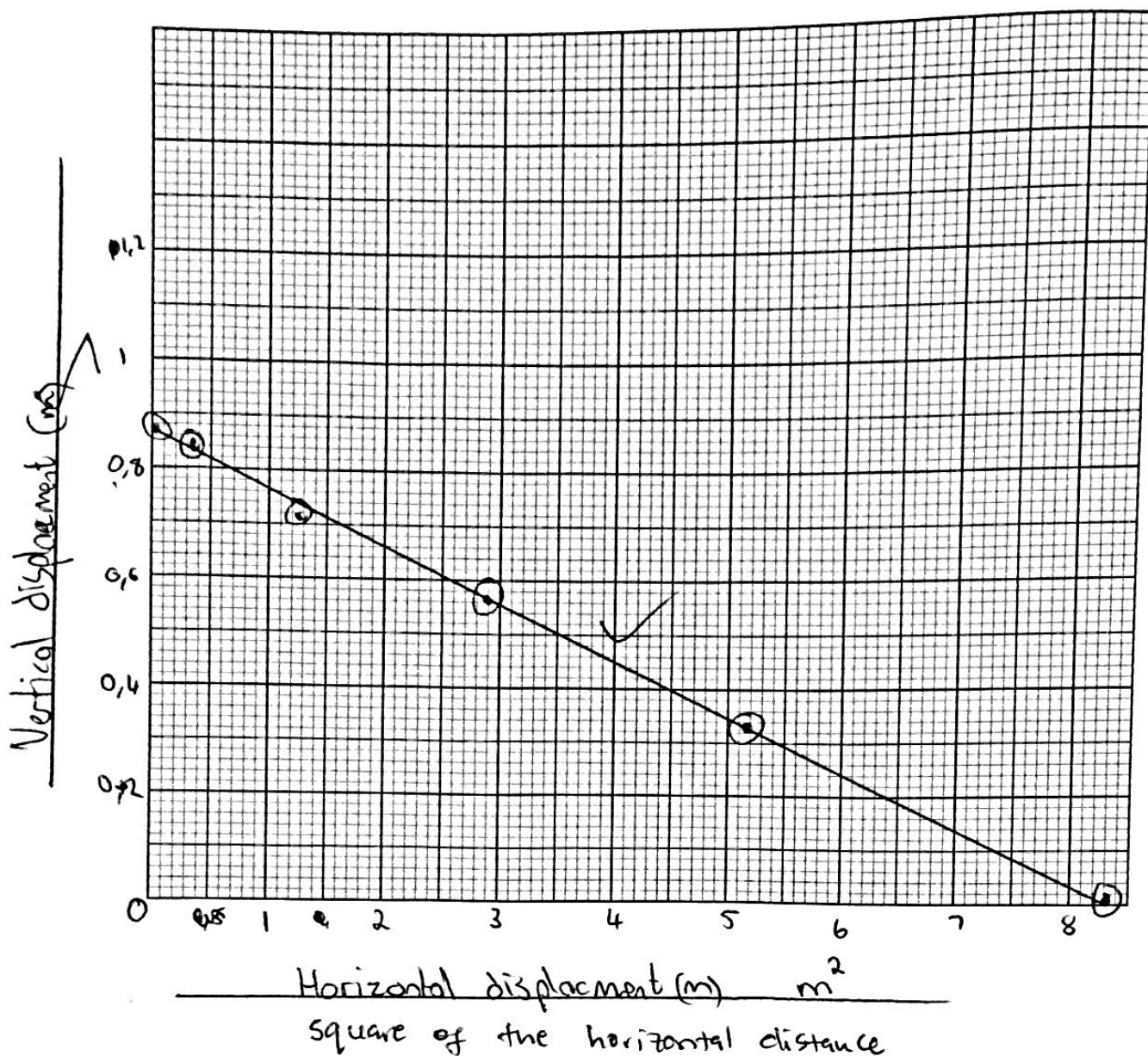
Conclusion

Viewing the results and values. It can be assumed that the actual values from the experiment are more accurate than the calculated.

DATE: 17 March

EKSP/ EXP: PAR

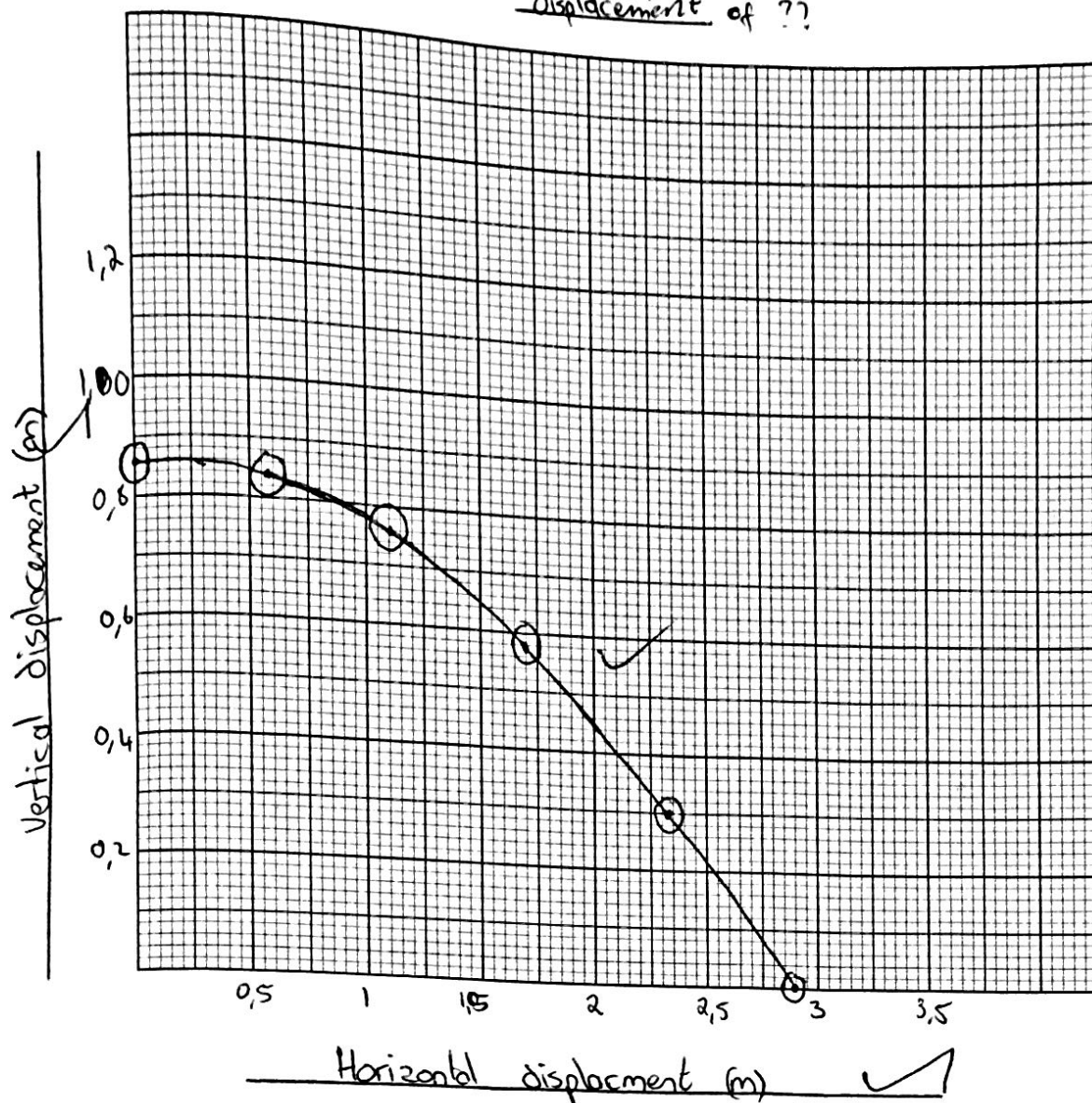
Graph showing vertical displacement as a function of horizontal displacement²



DATE: 17 Mar

EKSP/EXP: PAR

Graph showing vertical displacement as a function of Horizontal displacement of ??



12/05/15

Aim: To determine the relationship between the amount of heat added to a certain amount of water ^{or} the change in its temperature to determine the heat capacity.

mass of cup: 211g 0.0022kg

mass of water: 2105.6g 0.056kg

Initial temperature: ~~20.0~~ 294.15°K

Table showing the change in temperature of water brought about by the voltage and current of a resistor.

	T	$\Delta T(^{\circ}K)$	I (A)	V (V)	t (s)	Q (J)
	T_0	0.0	0	2.5	0	0
	T_1	0.8	2.5	3.5	60	625
	T_2	2.6	2.5	3.5	120	1050
	T_3	5.0	2.5	3.5	180	1575
	T_4	6.8	2.5	3.5	240	2160
	T_5	6.8	2.5	3.5	300	2625

$$\begin{aligned}
 4) \quad m &= \frac{\Delta y}{\Delta x} \\
 &= \frac{6.8 - 2.6}{2625 - 1050} \\
 &= 2.73 \times 10^{-3} \text{ units}^{-1}
 \end{aligned}$$

The relationship between the heat ^{added} and temperature of the water shows that it is directly proportional.

$$\begin{aligned}
 5) \quad y &= mx + c \\
 \Delta t &= (2.73 \times 10^{-3}) Q_w + 0 \text{ units}^{-1}
 \end{aligned}$$

$$6) \Delta t = \frac{1}{m_w C_w} Q_w$$

$$7) = \frac{1}{m_w C_w} = 2,73 \times 10^{-3}$$

$$8) \frac{1}{m_w C_w} = 2,73 \times 10^{-3}$$

$$\bullet \frac{1}{m_w (2,73 \times 10^{-3})} = C_w$$

$$\frac{1}{(0,105 \text{ kg}) (2,73 \times 10^{-3})} = C_w$$

$$C_w = 3488,5 \text{ J/kg} \cdot \text{K}$$

$$9) \frac{4200 - 3481,5}{4200} \times 100 = 17,9\% \text{ error}$$

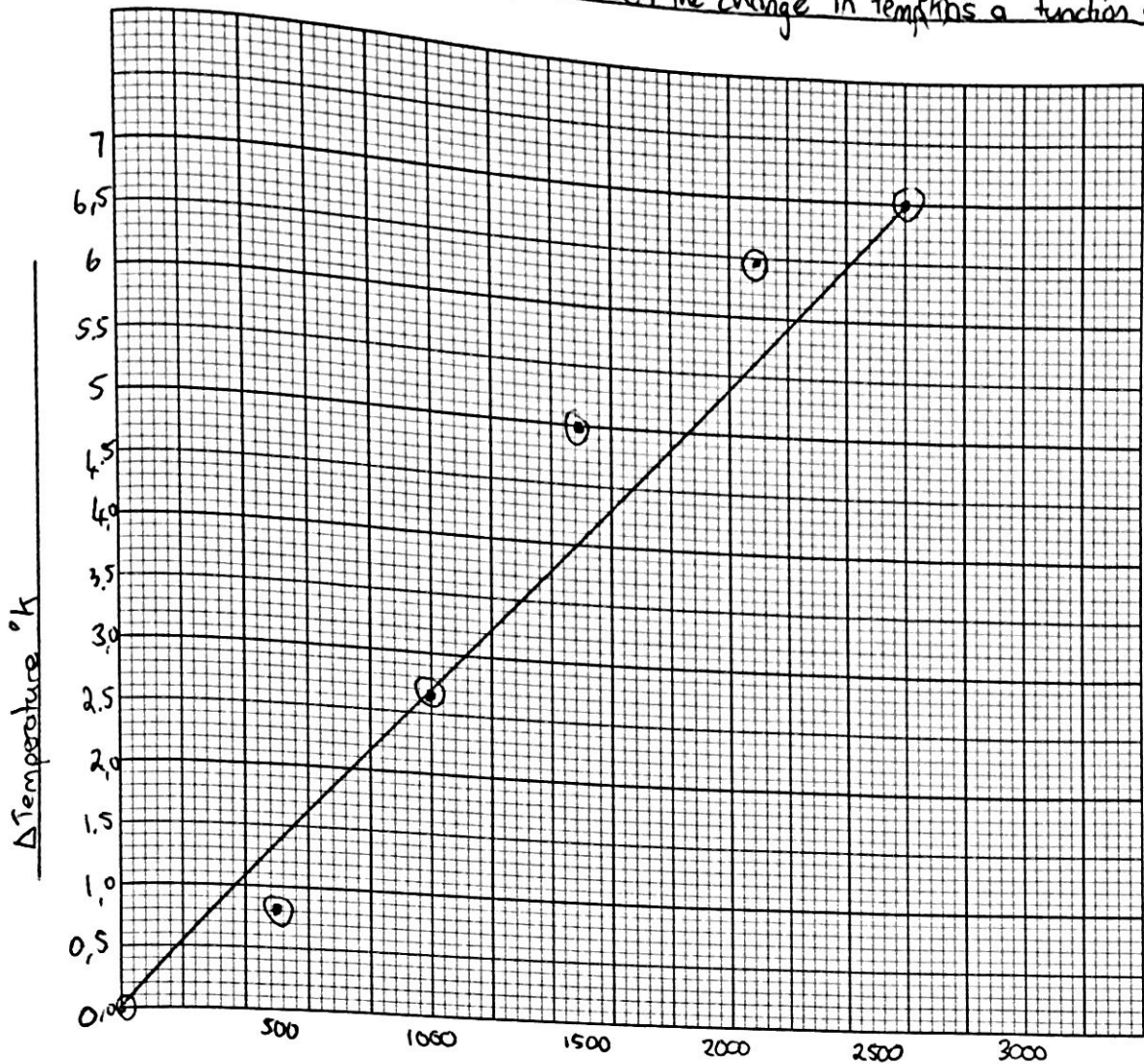
10) In accurate readings can result from interdependent ~~error~~ faulty equipment and the varying resistor might also contribute. This can all be stated as experimental error. Because the environment is not isolated ~~the~~ an error will also occur.

Conclusion.

DATE: 2015-05-12

EKSP/EXP: HEA.....

Graph showing relationship between the change in temperature a function of Heat (J)



Amount of HEAT (J)
Q_{in}

✓
②

2014-04-21

- ① Aim: To determine if the combined linear momentum of the disks was conserved in spite of the collision
- ② Table showing information regarding the collision of the objects.

	Before		After	
	Striker	Target	Striker	Target
Distance (m)	0,036	0	0,024	0,034
Time (s)	0,033	0	0,033	0,033
Velocity (m.s ⁻¹)	1,09 m.s ⁻¹	0	0,73 m.s ⁻¹	1,03 m.s ⁻¹
Mass (kg)	0,0145	0,0054	0,0145	0,0054
Angle (degree)	2,3°	0	-22°	37,6°
Momentum (kg.m/s)	0,0158	0	0,0105	0,0056

③ Calculations

V Before $V_s = \frac{\Delta s}{\Delta t} = \frac{0,036}{0,033} = 1,09 \text{ m.s}^{-1}$

$V_T = \frac{\Delta s}{\Delta t} = \frac{0}{0} = 0 \text{ m.s}^{-1}$

V After $V_s = \frac{\Delta s}{\Delta t} = \frac{0,24}{0,033} = 0,73 \text{ m.s}^{-1}$

$V_T = \frac{\Delta s}{\Delta t} = \frac{0,34}{0,33} = 1,03 \text{ m.s}^{-1}$

P Before $P_s = m \cdot v = (0,0145)(1,09) = 0,0158 \text{ kg.m.s}^{-1}$

$P_T = m \cdot v = (0,0054)(0) = 0 \text{ kg.m.s}^{-1}$

P After $P_s = m \cdot v = (0,0145)(0,73) = 0,0106 \text{ kg} \cdot \text{m} \cdot \text{s}^{-1}$

$P_T = m \cdot v = (0,0054)(1,03) = 0,0056 \text{ kg} \cdot \text{m} \cdot \text{s}^{-1}$

(4) Scale:

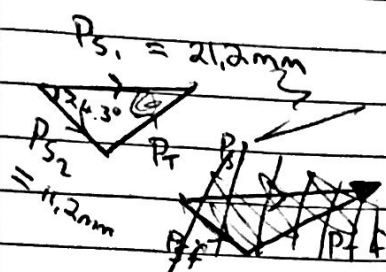
$0,01 \text{ kg} \cdot \text{m} \cdot \text{s}^{-1} = 20 \text{ mm}$

$0,0158 \text{ kg} \cdot \text{m} \cdot \text{s}^{-1} = 31,6 \text{ mm}$

$0,0106 \text{ kg} \cdot \text{m} \cdot \text{s}^{-1} = 21,2 \text{ mm}$

$0 \text{ kg} \cdot \text{m} \cdot \text{s}^{-1} = 0 \text{ mm}$

$0,0056 \text{ kg} \cdot \text{m} \cdot \text{s}^{-1} = 11,2 \text{ mm}$



$P_{T2} = 1,6 \text{ cm}$

$P_{T2} = M_T V_2$

$0,016 \text{ mm} = M_T (1,03)$

$M_2 = 0,0155 \text{ kg}$

measured mass = $0,0145 \text{ kg}$

Calculated mass = $0,0155 \text{ kg}$

$\% \text{ mass} = \frac{|m_{\text{scale}} - m_2|}{m_{\text{scale}}} \times 100 = \frac{|0,0145 - 0,0155|}{0,0145} \times 100 = 6,89\%$

Conclusion: The increased mass implies that the velocity was less and the velocity will be less due to friction, from the paper and many external forces. Experimental error can also be blamed. Therefore in conclusion linear momentum was ~~not~~ conserved and the change is experimental error of 6,89%.

14-04-2015

Aim: to determine if a spring-weight system follows the characteristics of simple harmonic motion.

① Determining k (spring)

Table showing the force, displacement & spring constant for a spring with a suspended weight.

Reading	Force (N)	Displacement (mm)	k (N/mm)
1	-1,862	-80mm	+ 0,0232
2	-0,916	-40mm	0,0229
3	-1,434	-60mm	0,0239
Avg	-1,404	-60mm	0,023

23.4 N/m

② Record several oscillations

table showing y displacements of oscillation.

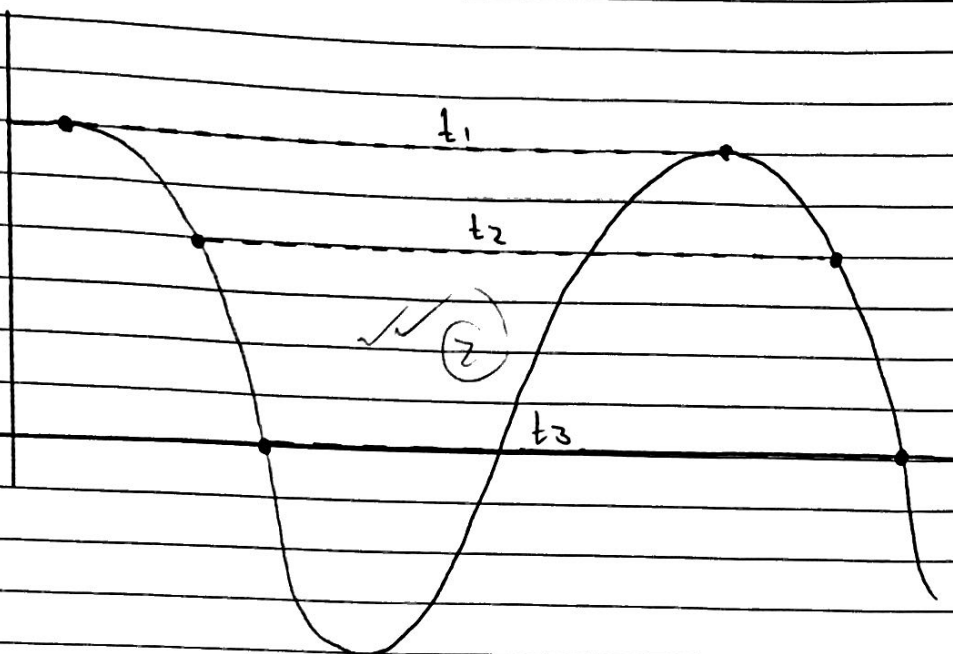
Reading	Time (s)	Force (N)	y_e
1	0,4556	0,980	42,6mm
2	0,5262	0,360	15,7mm
3	0,6573	-0,312	-13,57mm
4	0,7962	-0,985	-42,83mm
5	0,9585	-0,010	-0,43mm
6	1,1281	+0,973	42,30mm

①

Angular frequency + Amplitude

$$T_{ave} = \frac{0,6735s + 0,6696 + 0,6652 + \dots}{363}$$

$$= 0,6691s$$



Angular frequency

$$\text{Eq 5 : } \omega = \frac{2\pi}{T}$$

$$\text{Eq 8 : } \omega^2 = \frac{k}{m}$$

$$\text{Restoration force (max)} = 1N$$

$$\text{Restoration force (min)} = -1N$$

$$\text{Amplitude} = \frac{1N}{6,013N/m} = 43,48mm$$

5

$$\text{Eq 5} = \omega = 9.39 \text{ rad/s} \quad \checkmark$$

$$\begin{aligned} \text{Eq 8} = \omega^2 &= \frac{k}{m} \\ \omega &= \sqrt{\frac{k}{m}} \\ &= \sqrt{\frac{23.4 \text{ N/mm}}{0.250}} \end{aligned}$$

$$= 9.674 \text{ N/m} \cdot \text{kg} \quad \checkmark$$

$$\omega_{\text{avg}} = \frac{9.39 + 9.674}{2}$$

$$= 9.532 \text{ N/m} \cdot \text{kg} \quad \checkmark$$

Phase constant

$$\begin{aligned} \text{At } t=0 \quad F &= -0.410 \text{ N} \\ y &= -17.83 \text{ mm} \end{aligned}$$

$$\text{Eq 7} = y = y_m \cos(\omega t + \phi)$$

$$t=0 \quad ; \quad y_m = ?$$

$$\frac{-17.83}{43.48} = \cos \phi$$

$$\phi = \cos^{-1}\left(\frac{-17.83}{43.48}\right)$$

$$= 1.99 \text{ rad} \quad \checkmark$$



Reading	time	y_m	y	Δy
1				
2				
3				
4				

Conclusion

By looking at the information one can see that the spring has characteristics that are described by simple harmonic motion

24/03/15

EKLA

Aim: To investigate the elasticity of a steel spring and a rubber band copied

5)

Load (g)	Extension (mm)	Contraction (mm)	y'??
0	9	0	
50	20	20	
100	34	34	
150	47	47	
200	61	61	
250	74	74	

6)

Load (g)	Load Force (N)	
0	0	
50	0,489	
100	0,978	wrong units
150	1,467	
200	1,956	
250	2,445	

7) on graph paper GRA9

8) Yes it does obey because the extension is equal to contraction therefore it is at equilibrium and Hooke's law is obeyed.

9) $m = \frac{\Delta y}{\Delta x} = \frac{68-20}{2250-0,489}$

$= 27,61 \text{ mm/N}$

$y = 27,61x + 9$
 $= 27,61 \text{ mm/N} x + 9 \text{ mm}$

wrong units

$$10) \quad w = ky$$

$$\frac{w}{k} = y$$

$$11) \quad w = ky$$

$$k = \frac{w}{y}$$

$$= \frac{2,445 \text{ J}}{74-9 \text{ mm}}$$

$$k = 0,04 \frac{\text{J}}{\text{mm}}$$

$$12) \quad y - \text{extension}$$

$$= 23 \text{ mm}$$

$$m \cdot g = ky$$

$$m = \frac{ky}{g}$$

$$m = \frac{0,04 (23)}{9,81}$$

$$= 0,09 \text{ kg}$$

2) B)	mass (g)	Extension (mm)	Contraction (mm)	load force (J)
	0	22	28	0
	50	29	47	0,49
	100	43	72	0,98
	150	64	83	1,46
	200	88	98	1,96
	250	117	117	2,45

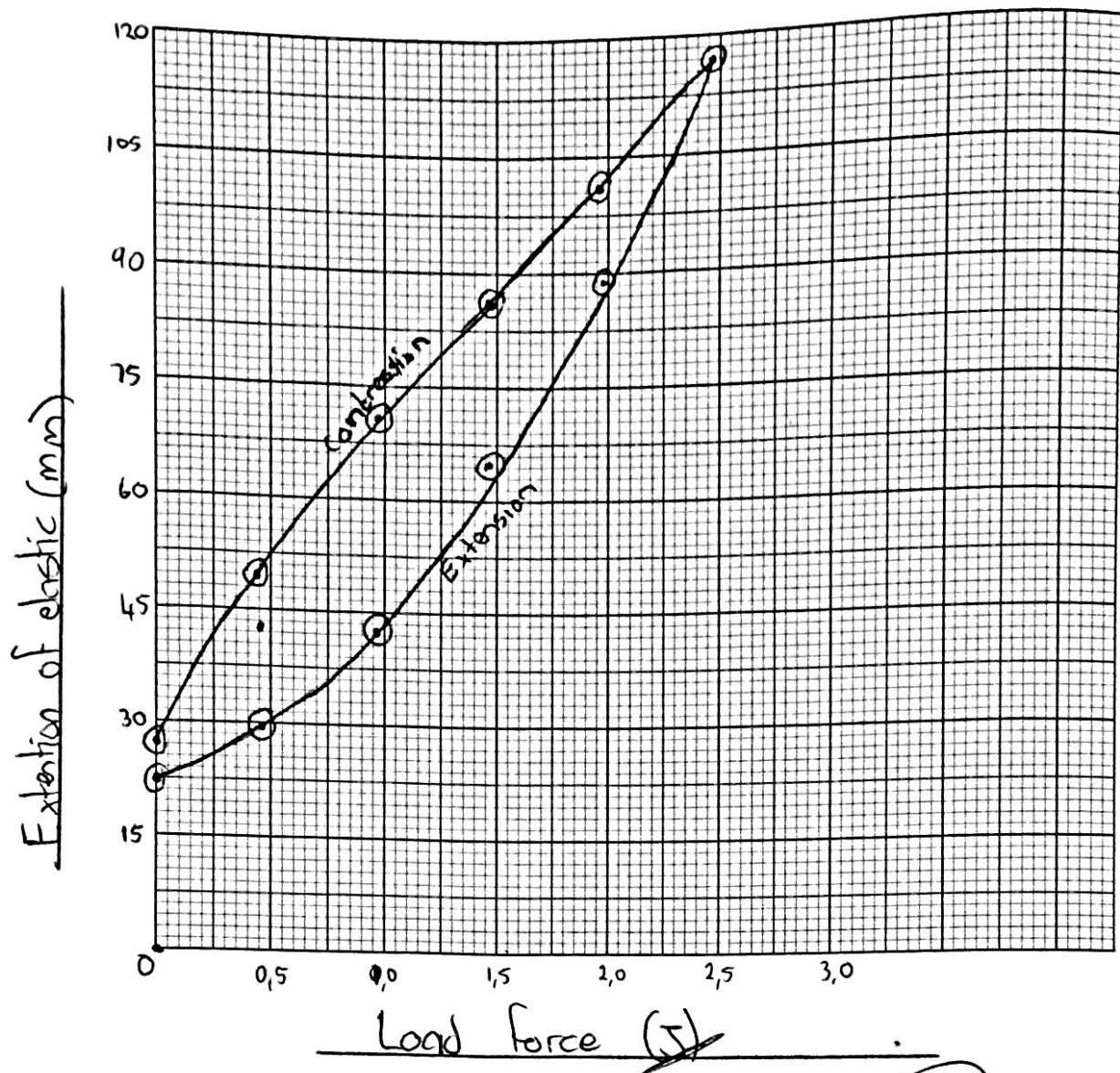
3) Graph on GRA 10

Conclusion : in this experiment it could be determined that a elastic band does not follow Hooke's law but a spring does.

DATE: 26 March

EKSP/ EXP: ELA

Extension of a elastic band as a function of load force.



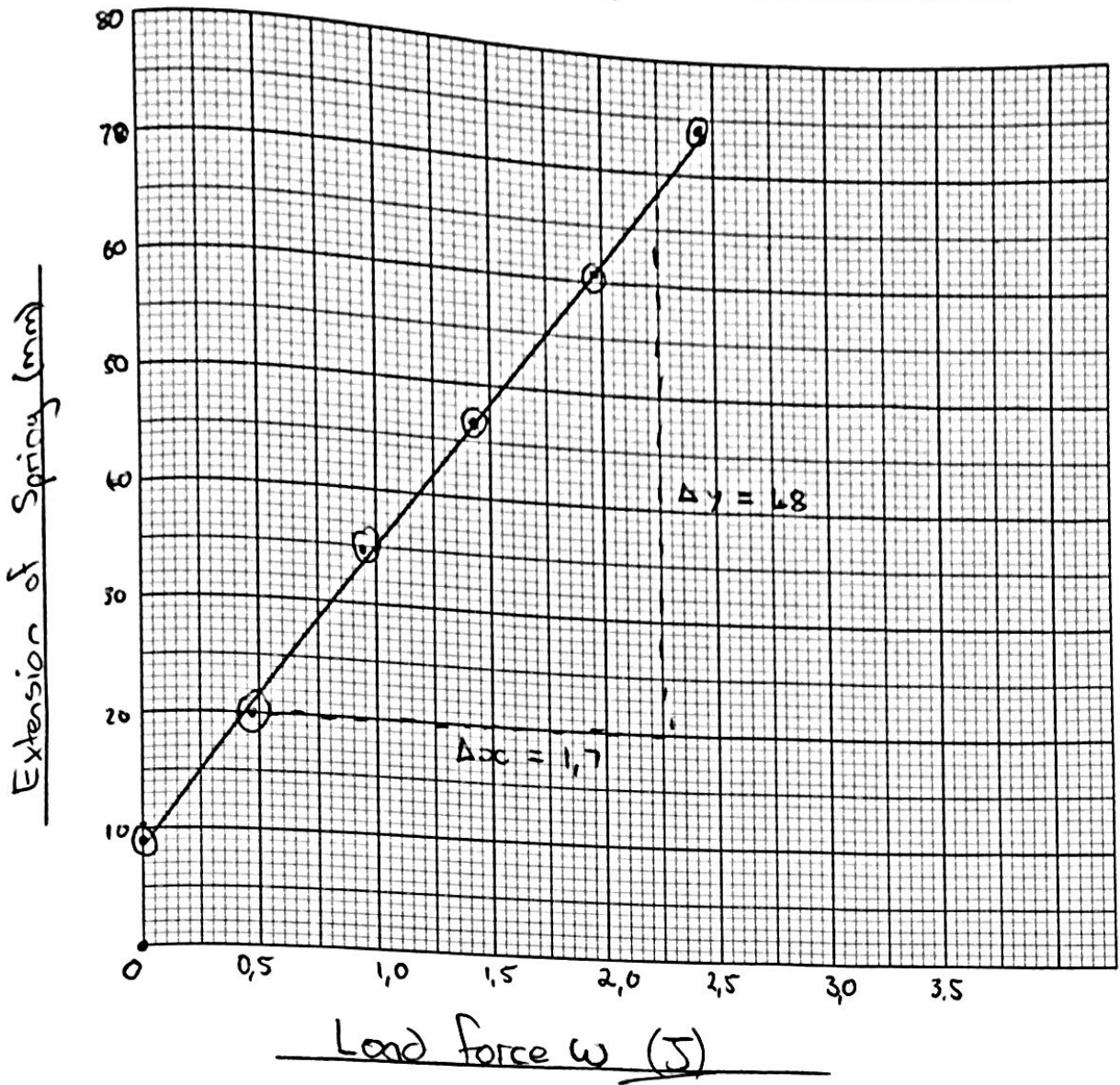
wrong units

ELA

DATE: 24/03/.....

EKSP/ EXP: ELA.....

Extension of the spring as a function of load force



②
wrong unit J